USN. | | | | | | | | | | [24MTCSE103]

FIRST SEMESTER M.Tech EXAMINATION, JANUARU- 2025

DISTRIBUTED COMPUTING

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Describe the key characteristics of distributed operating systems. How do these characteristics influence system performance and reliability?	CO 1	L2	**10**
	b.	Explain the client-server model in distributed computing. How does it facilitate resource sharing and communication in a network?	CO 1	L2	**10**
2.	**a.**	Discuss the CAP theorem and its implications for the design of distributed systems along with examples of systems that prioritize different trade-offs (e.g., CP, AP, or CA systems).	CO2	L2	**10**
	b.	Explain the importance of encryption and secure communication protocols in distributed systems along with commonly used encryption techniques and secure protocols like TLS/SSL.	CO2	L2	**10**
3.	**a.**	Explain the basic concept of virtualization and discuss its benefits for modern IT infrastructure, with emphasis on the role of the Hardware Abstraction Layer and Virtual Machine Monitor.	CO3	L2	**10**
	b.	Explain the concepts of Virtual LAN (VLAN) and Virtual Private Network (VPN) with their use cases.	CO3	L2	**10**
		OR			
4.	**a.**	Explain the components of the Security Model for virtual machines and its role in ensuring secure virtualization.	CO3	L2	**10**
	b.	How do mitigation models facilitate seamless resource and workload management across virtualized environments?	CO3	L3	**10**
5.	**a.**	As a cloud service consultant, how would you recommend the appropriate service delivery model (IaaS, PaaS, SaaS) for the following scenarios? **i.** Hosting a corporate website **ii.** Building a custom application	CO4	L3	**10**

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		iii. Managing customer relationship data			
	b.	Analyze the impact of adopting cloud service delivery and deployment models on business agility, cost efficiency, and scalability using relevant examples.	CO4	L4	**10**
		OR			
6.	**a.**	Consider a scenario where a multinational organization wants to migrate its IT infrastructure to the clouds. Which service delivery and deployment models would you recommend? Justify your choice based on scalability, cost, and security.	CO4	L3	**10**
	b.	Describe the role of the Hybrid Cloud deployment model in modern IT infrastructure. What are the key challenges in implementing a hybrid cloud environment, and how can they be addressed?	CO4	L3	**10**
7.	**a.**	Explain the importance of workload distribution and service load balancing in cloud architectures. How do these mechanisms enhance the reliability and performance of cloud services?	CO5	L2	**10**
	b.	Discuss cloud bursting and discuss its role in managing peak workloads. What are the challenges and advantages of implementing cloud bursting in hybrid cloud architectures?	CO5	L2	**10**
		OR			
8.	**a.**	Elaborate on the importance of dynamic failure detection and recovery in advanced cloud architectures.	CO5	L3	**10**
	b.	Describe the concept of hypervisor clustering and its role in achieving zero downtime	CO5	L2	**10**